

pCandidate seat No: _____

Charotar University of Science and Technology [CHARUSAT]
Faculty of Technology and Engineering
Department of Mathematical Sciences
MA243.01 TRANSFORMS AND VECTOR DIFFERENTIAL CALCULUS
2nd UNIT TEST

Semester: 3rd Sem. B. Tech. (EC)

Maximum Marks: 30

Date : 09/10/2019 (Wednesday)

Time: 09:15 am to 10:15 am

Instructions:

- (i) Figures to the right indicate **full** marks.
- (ii) Use of scientific calculator is allowed.
- (iii) Draw figure where it is required.

Q-1 Choose the correct answer from the given options in the following: [06]

1. Which of the following cannot be the Fourier series expansion of a periodic function?
(a) $f(x) = 2 \cos x + 3 \cos 3x$ (b) $f(x) = 2 \cos \pi x + 7 \cos x$
(c) $f(x) = \cos x + 0.5$ (d) $f(x) = 2 \cos 1.5 \pi x + 7 \cos 3.5 \pi x$
2. If from the function $f(t)$ one forms the function $\psi(t) = f(t) + f(-t)$, then $\psi(t)$ is _____.
(a) *even* (b) *odd*
(c) *neither even nor odd* (d) *both even and odd*
3. The Fourier transform of a function $f(x)$ is $F(s)$. The Fourier transform of $f'(x)$ will be _____.
(a) $\frac{dF(s)}{ds}$ (b) $i2\pi sF(s)$
(c) $isF(s)$ (d) $\frac{F(s)}{is}$
4. The value of $\int_0^\infty \frac{\sin^2 x}{x^2} dx$ is _____.
(a) $\frac{3\pi}{2}$ (b) $\frac{\pi^2}{2}$
(c) $\frac{\pi^2}{4}$ (d) $\frac{\pi}{2}$
5. If $U(z)$ is the Z-transform of signal u_n , then the Z-transform of $\delta(n)$ is _____.
(a) 1 (b) 0
(c) -1 (d) *not possible*
6. The Z-transform of the signal u_k is _____.
(a) $\sum_{k=0}^{\infty} u_k z^k$ (b) $\sum_{k=0}^{\infty} u_{-k} z^k$
(c) $\sum_{k=0}^{\infty} u_k z^{-k}$ (d) $\sum_{k=0}^{\infty} u_{-k} z^{-k}$

Q-2 Attempt any Three.

[12]

- (a) I. Find the Laplace transform of $e^t \cdot u(t - 2)$.
II. Solve $y'' + y = \delta(t - 3)$ with $y(0) = 0$ and $y'(0) = 1$.
- (b) Solve $y'' + 2y' + 5y = e^{-t} \sin t$ with $y(0) = 0$ and $y'(0) = 1$.
- (c) Find the Fourier series of $f(x) = \sqrt{1 - \cos x}$ in the interval $(0, 2\pi)$. Hence, deduce that $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}$.
- (d) Find the Fourier sine series of $f(x) = e^x$ in $0 < x < \pi$ and also find the Fourier cosine series of $f(x) = e^{-x}$, where $0 \leq x \leq \pi$.

Q-3 Attempt any Three.

[12]

- (a) Find the Fourier sine and cosine transform of $f(x) = e^{-ax}$.
- (b) Find $f(x)$ if the Fourier sine transform of it is $\frac{s}{s^2 + 1}$.
- (c) Find the Z-transform of a sequence $\frac{1}{n!}$ and hence find
 $Z\left[\frac{1}{(n-1)!}\right], Z\left[\frac{1}{(n+1)!}\right], Z\left[\frac{1}{(n+2)!}\right]$.
- (d) Find the inverse Z-transform of $\frac{2z^2 + 3z}{(z-4)(z+2)}$.
